COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
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ACTIVITY NO. 14.1 Algorithm Complexity

1. PROCEDURE OUTPUT

What procedures did you perform?

	Bubble Sort	Quick Sort
Best Case	O(n)	O(n log n)
Worst Case	O(n^2)	O(n^2)
Analysis	O(n) in linear time and O(n^2) quadratic time, O(n) the numbers are already sorted,however O(n^2) is the worst case because it repeats the process multiple times.	O(n log n) because of the balanced partitioning of sub arrays.

Table 14-1. Best- and Worst-Case Analysis using Theoretical Tools

	Bubble Sort	Quick Sort
Best Case	`Best case: 0.001 ms`	`Best case: 0.0109 ms`
Worst Case	`Worst case: 0.0891 ms`	`Worst case: 0.0735 ms`
Analysis	Best case is only 0.001 ms execution time. Means bubble sort quickly finishes and data already sorted.worst case took longer 0.0891 ms. slower because bubble sort needs more comparison.	It took 0.0735 ms because of the unbalanced splits requiring more sub array steps

Table 14-2. Best- and Worst-Case Analysis using Experimental Tools

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO A: Contrast and compare the two searching techniques used in class. The required deliverable for this activity would be the theoretical and empirical analysis. This is the following result of table and later i give the summary,

Linear search checks items one by one, so its work grows directly with the list size O(n), causing its running time to increase about ten times whenever the data grows ten times, moving from 0.0063 ms for 1,000 items up to 0.6106 for 100,000 items. In contrast, binary search repeatedly cuts a sorted list in half, so its total work grows very slowly 0(log N). Increasing the list size from 1,000 to 100,000 items adds only 7 extra comparison steps,which takes so little time that the recorded result stays fixed at 0.003ms across all tests. Ultimately, binary search scales much better for large amounts of data, though it requires the list to be sorted before searching.

Linear Search takes O(N) time complexity because the processor checks every item one after another, and that causes execution time to increase from 0.0063 ms to 0.6106 ms as the set ranges from 1,000 to 100,000 items while maintaining O(1) by searching directly in place. Then binary search takes O(log N) time complexity by dividing the set in half each step, then adding seven comparisons across that same range to hold the execution time constant at 0.0003 ms.

Input Size (N)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1,000	Binary Search (Worst-Case): 0.0003 ms	Linear Search (Worst-Case): 0.0063 ms
10,000	Binary Search (Worst-Case): 0.0003 ms	Linear Search (Worst-Case): 0.0609 ms
100,000	Binary Search (Worst-Case): 0.0003 ms	Linear Search (Worst-Case): 0.6106 ms

Table 14-3: Experimental Analysis of Worst time complexity of each algorithm

The linear search scales up because the program looks through the items one by one. And bigger lists mean more work, which shows clearly in the timing results. While binary search cuts the list in half over and over. And even on huge lists, the computer finishes those few extra checks in nanoseconds, so the timer numbers barely even change. Both methods searches inside directly the original array and that saves memory by not creating new ones.

Conclusion:
Formalejo: In this activity, I learned why binary search is faster and consistent compared to linear search because it cuts the list in half during each step. It keeps the execution time consistent at 0.0003 ms up to 100,000 items. While linear search slows down directly as list sizes increase, jumping from 0.0063 ms to 0.6106 ms. I also learned that quick sort handles worst case things faster than bubble sort. Overall, this one is a 10/10, not that hard because I had my other activities as a guide for this activity.

Licas:
For my conclusion, Based what part i made in this activity is all about the based and worst case about O(n) and O(n^2), I'd learn how can be a O(n) best case scenario compared to the worst case scenario of O(n^2), the 0(n) is a linear time and other one is on the quadratic time. Best case because it's already in the ascending order compared to the first one with so many switches and processes that you need to do for the worst case scenario. Pseudocode analysis, this part will learn about analysis of how the binary and linear in their best case scenario, the time doubles in 10X.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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